

BST Physics Curriculum Vol 12 Ch 4 — Chemical Reactions + Thermodynamics v0.4 (Calibration #23 substance-floor refill)

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redirect)

Vol 12 Chapter 4 — Chemical Reactions + Thermodynamics

Chapter motivation

Chemical reactions transform reactants to products under thermodynamic + kinetic control. The Gibbs free energy $\Delta G = \Delta H - T\Delta S$ determines spontaneity ($\Delta G < 0$ = spontaneous); the reaction rate determines how fast (Vol 12 Ch 5 kinetics). Standard chemistry treatment: enthalpy (Hess's law, bond enthalpies, heats of formation/combustion), entropy (Third Law, Boltzmann $S = k \ln W$), Gibbs free energy, equilibrium constants K_{eq} , Le Chatelier's principle, coupled reactions in biochemistry (ATP hydrolysis $\Delta G \approx -30$ kJ/mol). BST substrate-cartography reading: thermodynamic state functions (H, S, G) are substrate Zone-2/Zone-3 commitment-cycle bookkeeping quantities (T2417 4-Zone Commitment Cycle). Cross-link to Vol 6 Thermo + Stat Mech for general thermodynamic framework.

Section 4.0b — Reader-grade 3-level pedagogy

Level 1: Reactions go forward when $\Delta G = \Delta H - T\Delta S < 0$. Enthalpy = bond energy bookkeeping (Zone-2 substrate commitment); entropy = configuration counting (Zone-3 substrate emission); Gibbs free energy combines them. Chemical equilibrium = substrate commitment-emission balance.

Level 2 (graduate-chemist accessible): Enthalpy ΔH (heat content): Hess's law states ΔH for reaction equals sum of bond breaking minus bond forming enthalpies (or equivalently, products' heats of formation minus reactants'). Standard heats of formation tabulated for thousands of compounds. Bond enthalpies (C-H ~413 kJ/mol, C=C ~614 kJ/mol, C≡C ~839 kJ/mol). Combustion enthalpies (glucose $C_6H_{12}O_6 + 6O_2 \rightarrow 6CO_2 + 6H_2O$ has $\Delta H \approx -2802$ kJ/mol). Entropy ΔS : Boltzmann

$S = k_B \ln W$ where W is microstate count; Third Law $S \rightarrow 0$ as $T \rightarrow 0$ for perfect crystal. Standard molar entropies tabulated ($S^\circ(\text{gases}) > S^\circ(\text{liquids}) > S^\circ(\text{solids})$). Gibbs free energy $\Delta G = \Delta H - T\Delta S$: $\Delta G < 0$ spontaneous; $\Delta G = 0$ equilibrium; $\Delta G > 0$ non-spontaneous. Equilibrium constant $K_{eq} = \exp(-\Delta G^\circ/RT)$; reaction quotient Q vs K_{eq} determines reaction direction. Le Chatelier's principle: equilibrium shifts to counteract applied stress (concentration, temperature, pressure). Coupled reactions (ATP hydrolysis $\Delta G \approx -30$ kJ/mol, drives non-spontaneous biochemistry). BST substrate-cartography reading per T2417 4-Zone Commitment Cycle: Zone-1 = K-type generation (substrate absorption); Zone-2 = commitment processing (chemical bond formation/breaking, ΔH bookkeeping); Zone-3 = emission residue (entropy ΔS bookkeeping as substrate microstate counts); Zone-4 = outer-edge bookkeeping (Λ scale, irrelevant to chemistry but provides the thermodynamic temperature scale via cosmological substrate). Gibbs free energy = Zone-2 + Zone-3 net substrate balance. Chemical equilibrium = substrate commitment-emission cycle balance at given T . Cross-link to Vol 6 Ch 5 (Helmholtz + Gibbs free energies as Legendre transforms of internal energy).

Level 3 (5th-grader accessible): Reactions happen when they “want to” — when the products are more stable than the reactants. Two things matter: energy (do bonds get stronger?) and disorder (do things spread out more?). Gibbs free energy combines them: $\Delta G = \Delta H - T\Delta S$. If ΔG is negative, the reaction goes forward. Equilibrium is when forward and backward rates balance. BST framework says these energy + disorder quantities are bookkeeping for substrate processing cycles (Zones 2 + 3 of the 4-Zone substrate cycle).

Section 4.1 — Enthalpy and Hess's Law

Hess's law: ΔH for any chemical reaction equals the sum of ΔH for individual reaction steps along any pathway. Equivalent: $\Delta H_{rxn} = \sum \nu_{products} \cdot \Delta H^\circ_f(\text{products}) - \sum \nu_{reactants} \cdot \Delta H^\circ_f(\text{reactants})$.

Bond enthalpies estimate: $\Delta H_{rxn} \approx \Sigma(\text{bonds broken}) - \Sigma(\text{bonds formed})$. Approximation since bond enthalpies are averages over many molecules.

Standard combustion enthalpies tabulated for fuels (methane ~ 890 , propane ~ 2220 , glucose ~ 2802 kJ/mol).

BST framing: ΔH = Zone-2 substrate commitment-bookkeeping. Bond enthalpies = substrate K-type Casimir-eigenvalue differences (Vol 12 Ch 2 cross-link).

Section 4.2 — Entropy and Third Law

Boltzmann entropy: $S = k_B \ln W$ where W is microstate count. Third Law: $S \rightarrow 0$ as $T \rightarrow 0$ K for perfect crystal (only one microstate). Standard molar entropies tabulated; gases $\gg$ liquids $>$ solids. Phase transitions: $\Delta S_{vap} > 0$ (large), $\Delta S_{fus} > 0$ (small).

BST framing: ΔS = Zone-3 substrate emission-bookkeeping (microstate counting at substrate sub-structure level).

Section 4.3 — Gibbs Free Energy and Equilibrium

$\Delta G = \Delta H - T\Delta S$. Spontaneity criterion: $\Delta G < 0$ at constant T, P.

Four cases by signs of ΔH and ΔS : | ΔH | ΔS | Spontaneous? | |---|---|---|---| |
 - | + | Always | | - | - | Low T only | | + | + | High T only | | + | - | Never |

Equilibrium constant: $K_{eq} = \exp(-\Delta G^\circ/RT)$. At equilibrium, $Q = K_{eq}$.

Le Chatelier's principle: equilibrium shifts to counteract stress (add reactant → shifts toward products; raise T for endothermic → shifts toward products).

BST framing: equilibrium = substrate commitment-emission balance at given T.

Section 4.4 — Coupled Reactions (Biochemistry Bridge)

Coupled reactions allow non-spontaneous reactions to proceed by coupling to spontaneous ones. ATP hydrolysis ($ATP + H_2O \rightarrow ADP + Pi$) has $\Delta G^\circ \approx -30$ kJ/mol; couples to many biosynthetic reactions in metabolism (Vol 13 Ch 9 metabolism cross-link).

BST framing: coupled reactions = substrate commitment-cycle linking; energy currency molecules (ATP, NADH, $FADH_2$) are substrate Zone-2/Zone-3 commitment carriers.

Section 4.5 — Connection

- Vol 6 Ch 5+6+7 Thermodynamic potentials (general framework)
- Vol 12 Ch 2-3 Chemical bonding + MO theory (bond enthalpy origin)
- Vol 12 Ch 5 Reaction Kinetics + Transition States (kinetics complements thermo)
- Vol 13 Ch 9 Metabolism (coupled reactions in biology)
- T2417 4-Zone Commitment Cycle (substrate bookkeeping framework)

Honest scope: T2417 4-Zone framework provides organizing principle for substrate-bookkeeping interpretation; specific quantitative reaction free-energy derivations from substrate Hamiltonian remain multi-year SP-31 program. ATP coupling ~ -30 kJ/mol is empirically tabulated value, not yet substrate-derived.

— Lyra, Vol 12 Ch 4 v0.4 chapter-grade narrative REFILLED per Cal #104 + Calibration #23, Saturday 2026-05-23 EDT